5/3/2026

Project Report

Financial Reconciliation System (AMF)

Vijay Bandewar

Project Report

Project Title: Automated Multi-Source Financial Reconciliation System (AMF)
Submitted By: Vijay
Date: May 2026

Abstract

Daily financial reconciliation requires comparing transactions received from external sources (commonly Excel files) against internal system records stored in a database. Manual reconciliation is time consuming, error prone, and delays operational reporting. This project implements an automated reconciliation pipeline that ingests external Excel transactions using Python and Pandas, stores canonical data in Oracle, applies reconciliation rules via PL/SQL stored procedures, logs matches and discrepancies into audit tables, and generates stakeholder ready Excel reports with dashboards and conditional formatting for high value discrepancy alerts.

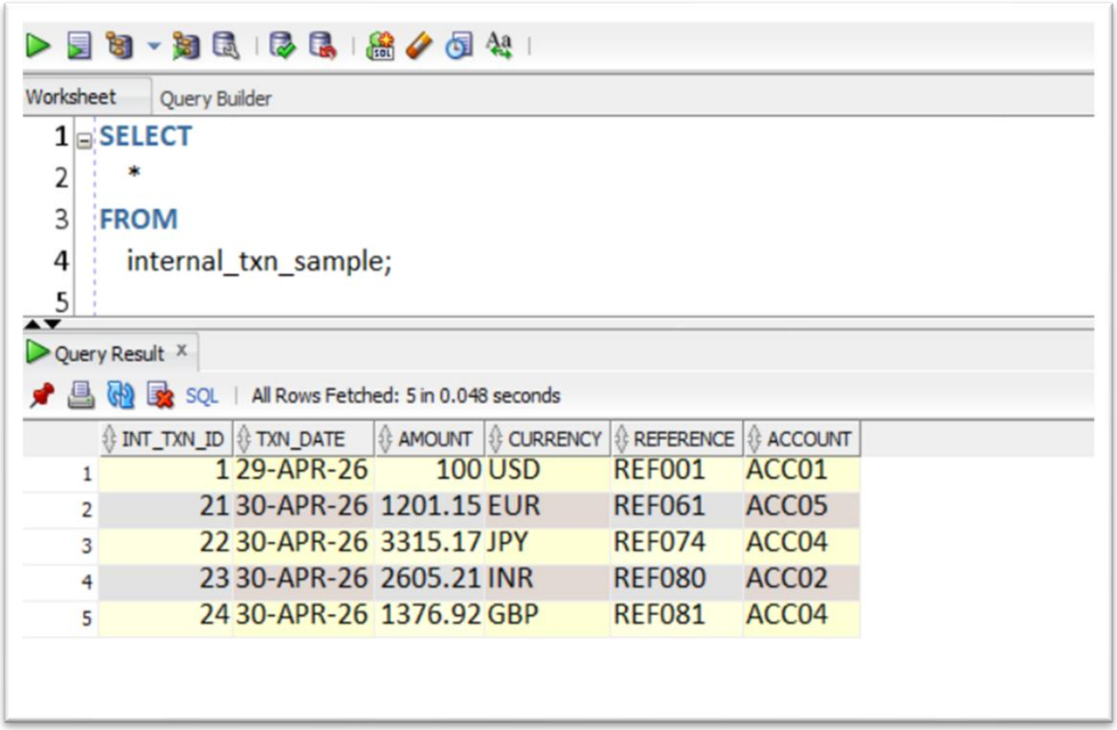
Table of Contents

Project Trailer (Demo Snaps):

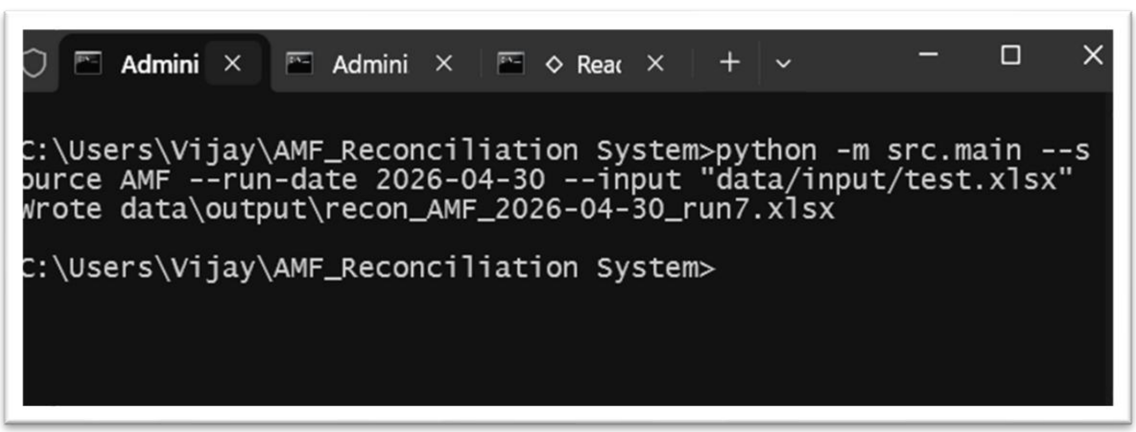
This section is meant to “show what the project does” at a glance. Paste screenshots into docs/images/ and the images will display inside this report.

Recommended screenshots

S3 —Input File Details:
The required columns for the input file, test.xlsx, are: txn_date, amount, currency, reference, and account.

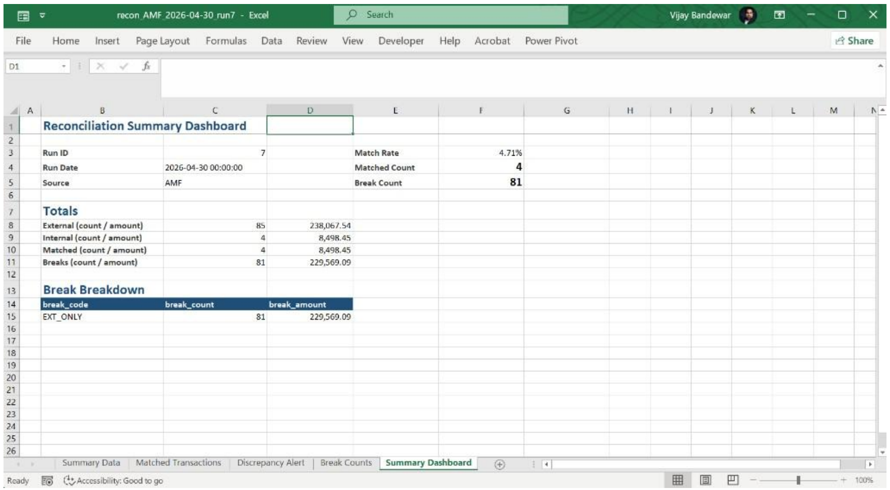


S3 —Command Execution:
The PowerShell output, specifically the execution of python -m src.main ..., generates a report, the filename of which is also displayed.

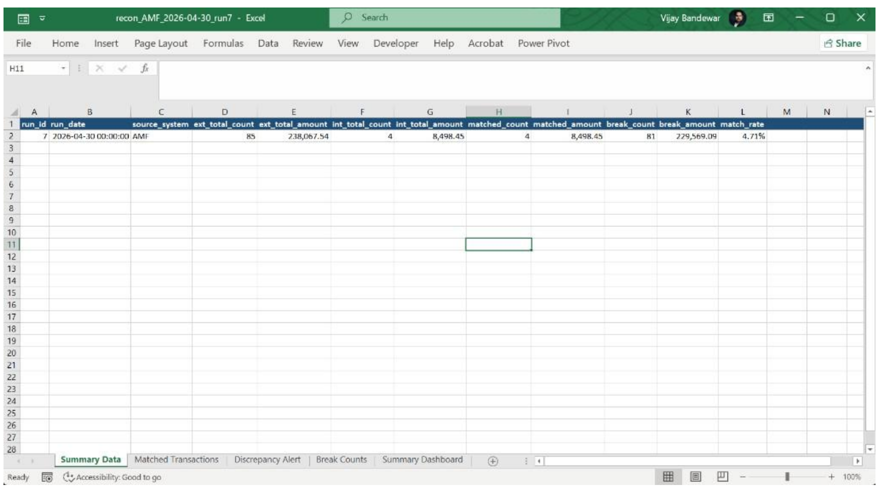


S3 — Output Workbook Tabs:

A screenshot of the output Excel file, and the tabs (Summary Dashboard, Matched Transactions, and Discrepancy Alert) are visible at the bottom.



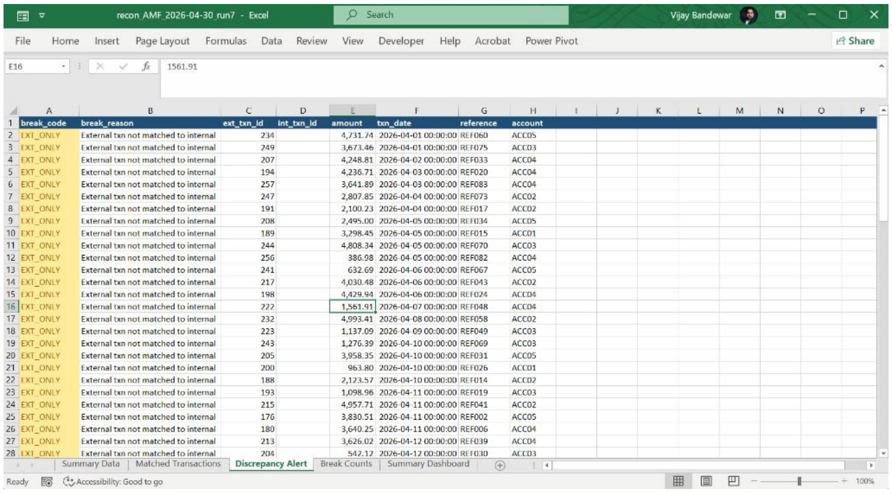
Reconciliation Summary Dashboard				
Run ID	7		Match Rate	4.71%
Run Date	2026-04-30 00:00:00		Matched Count	4
Source	AMF		Break Count	81
Totals				
External (count / amount)	85	238,067.54		
Internal (count / amount)	4	8,498.45		
Matched (count / amount)	4	8,498.45		
Breaks (count / amount)	81	229,569.09		
Break Breakdown				
break_code	break_count	break_amount		
EXT_ONLY	81	229,569.09		



run_id	run_date	source_system	ext_total_count	ext_total_amount	int_total_count	int_total_amount	matched_count	matched_amount	break_count	break_amount	match_rate
7	2026-04-30 00:00:00	AMF	85	238,067.54	4	8,498.45	4	8,498.45	81	229,569.09	4.71%

S3 — Discrepancy Highlight:

Include a screenshot demonstrating the use of conditional formatting to emphasize a high-value break.



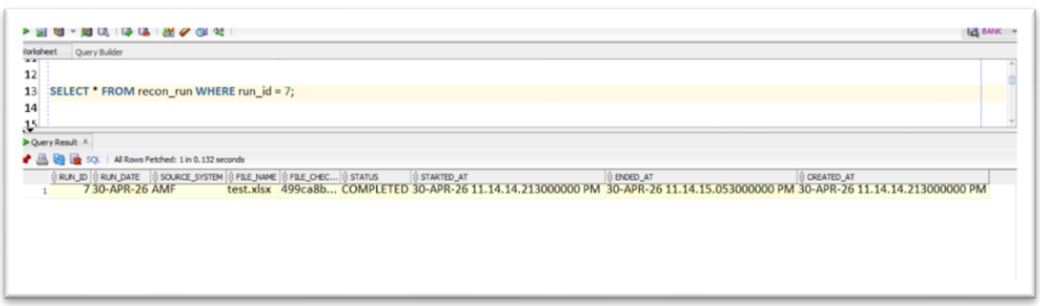
recon_AMF_2026-04-30_run7 - Excel

	break_code	break_reason	ext_txn_id	int_txn_id	amount	txn_date	reference	account
2	EXT_ONLY	External txn not matched to internal	234		4,731.74	2026-04-01 00:00:00	REF060	ACC05
3	EXT_ONLY	External txn not matched to internal	249		3,673.46	2026-04-01 00:00:00	REF075	ACC03
4	EXT_ONLY	External txn not matched to internal	207		4,248.81	2026-04-02 00:00:00	REF033	ACC04
5	EXT_ONLY	External txn not matched to internal	194		4,236.71	2026-04-03 00:00:00	REF020	ACC04
6	EXT_ONLY	External txn not matched to internal	257		3,641.89	2026-04-03 00:00:00	REF083	ACC04
7	EXT_ONLY	External txn not matched to internal	247		2,807.85	2026-04-04 00:00:00	REF073	ACC02
8	EXT_ONLY	External txn not matched to internal	191		2,100.23	2026-04-04 00:00:00	REF017	ACC02
9	EXT_ONLY	External txn not matched to internal	208		2,495.00	2026-04-05 00:00:00	REF034	ACC05
10	EXT_ONLY	External txn not matched to internal	189		3,298.45	2026-04-05 00:00:00	REF015	ACC01
11	EXT_ONLY	External txn not matched to internal	244		4,808.34	2026-04-05 00:00:00	REF070	ACC03
12	EXT_ONLY	External txn not matched to internal	256		386.98	2026-04-05 00:00:00	REF082	ACC04
13	EXT_ONLY	External txn not matched to internal	241		632.69	2026-04-06 00:00:00	REF067	ACC05
14	EXT_ONLY	External txn not matched to internal	217		4,030.48	2026-04-06 00:00:00	REF043	ACC02
15	EXT_ONLY	External txn not matched to internal	198		4,429.94	2026-04-06 00:00:00	REF024	ACC04
16	EXT_ONLY	External txn not matched to internal	222		1,561.91	2026-04-07 00:00:00	REF048	ACC04
17	EXT_ONLY	External txn not matched to internal	232		4,993.41	2026-04-08 00:00:00	REF058	ACC02
18	EXT_ONLY	External txn not matched to internal	223		1,137.09	2026-04-09 00:00:00	REF049	ACC03
19	EXT_ONLY	External txn not matched to internal	243		1,276.39	2026-04-10 00:00:00	REF069	ACC03
20	EXT_ONLY	External txn not matched to internal	205		3,958.35	2026-04-10 00:00:00	REF031	ACC05
21	EXT_ONLY	External txn not matched to internal	200		963.80	2026-04-10 00:00:00	REF026	ACC01
22	EXT_ONLY	External txn not matched to internal	188		2,123.57	2026-04-10 00:00:00	REF014	ACC02
23	EXT_ONLY	External txn not matched to internal	193		1,098.96	2026-04-11 00:00:00	REF019	ACC03
24	EXT_ONLY	External txn not matched to internal	215		4,957.71	2026-04-11 00:00:00	REF041	ACC02
25	EXT_ONLY	External txn not matched to internal	176		3,830.51	2026-04-11 00:00:00	REF002	ACC05
26	EXT_ONLY	External txn not matched to internal	180		3,640.25	2026-04-11 00:00:00	REF006	ACC04
27	EXT_ONLY	External txn not matched to internal	213		3,626.02	2026-04-12 00:00:00	REF039	ACC04
28	EXT_ONLY	External txn not matched to internal	204		542.12	2026-04-12 00:00:00	REF030	ACC03

Summary Data | Matched Transactions | Discrepancy Alert | Break Counts | Summary Dashboard

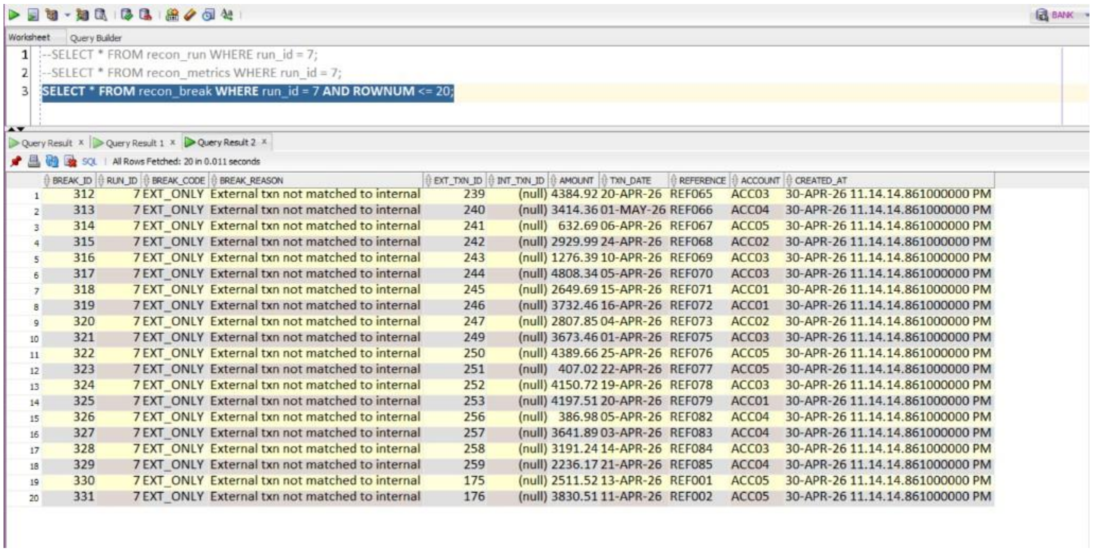
S5 — Oracle Audit Tables:

Include a screenshot of the recon_run, recon_metrics, and recon_break tables from the Oracle audit database for a specific run_id.



	RUN_ID	RUN_DATE	SOURCE_SYSTEM	FILE_NAME	FILE_CHEC...	STATUS	STARTED_AT	ENDED_AT	CREATED_AT
1	7	30-APR-26	AMF	test.xlsx	499ca8b...	COMPLETED	30-APR-26 11.14.14.213000000 PM	30-APR-26 11.14.15.053000000 PM	30-APR-26 11.14.14.213000000 PM

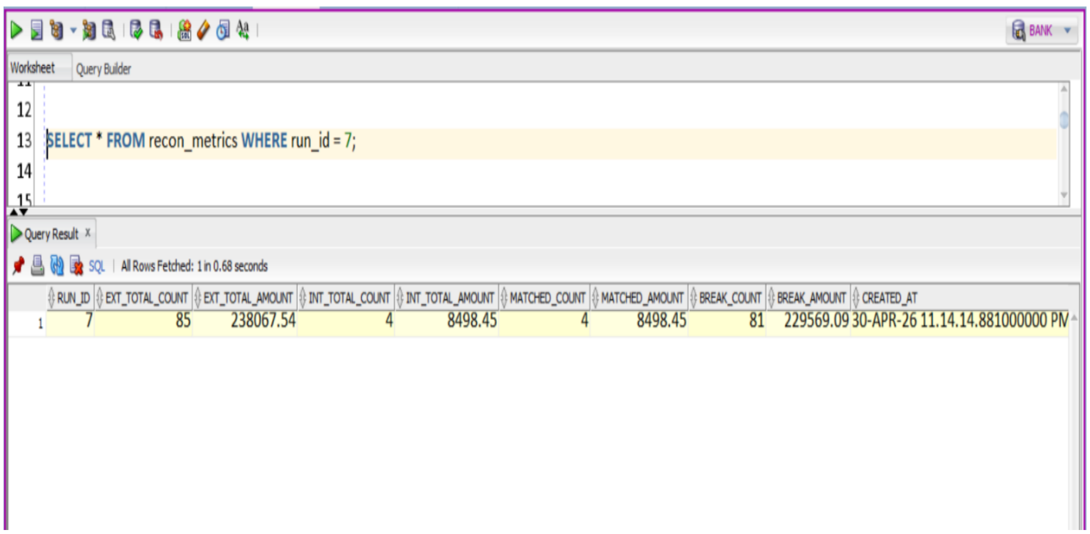
	RUN_ID	EXT_TOTAL_COUNT	EXT_TOTAL_AMOUNT	INT_TOTAL_COUNT	INT_TOTAL_AMOUNT	MATCHED_COUNT	MATCHED_AMOUNT	BREAK_COUNT	BREAK_AMOUNT	CREATED_AT
1	7	85	238067.54	4	8498.45	4	8498.45	81	229569.09	30-APR-26 11.14.14.881000000 PM



```
--SELECT * FROM recon_run WHERE run_id = 7;
--SELECT * FROM recon_metrics WHERE run_id = 7;
SELECT * FROM recon_break WHERE run_id = 7 AND ROWNUM <= 20;
```

All Rows Fetched: 20 in 0.011 seconds

	BREAK_ID	RUN_ID	BREAK_CODE	BREAK_REASON	EXT_TXN_ID	INT_TXN_ID	AMOUNT	TXN_DATE	REFERENCE	ACCOUNT	CREATED_AT
1	312	7	EXT_ONLY	External txn not matched to internal	239	(null)	4384.92	20-APR-26	REF065	ACC03	30-APR-26 11.14.14.861000000 PM
2	313	7	EXT_ONLY	External txn not matched to internal	240	(null)	3414.36	01-MAY-26	REF066	ACC04	30-APR-26 11.14.14.861000000 PM
3	314	7	EXT_ONLY	External txn not matched to internal	241	(null)	632.69	06-APR-26	REF067	ACC05	30-APR-26 11.14.14.861000000 PM
4	315	7	EXT_ONLY	External txn not matched to internal	242	(null)	2929.99	24-APR-26	REF068	ACC02	30-APR-26 11.14.14.861000000 PM
5	316	7	EXT_ONLY	External txn not matched to internal	243	(null)	1276.39	10-APR-26	REF069	ACC03	30-APR-26 11.14.14.861000000 PM
6	317	7	EXT_ONLY	External txn not matched to internal	244	(null)	4808.34	05-APR-26	REF070	ACC03	30-APR-26 11.14.14.861000000 PM
7	318	7	EXT_ONLY	External txn not matched to internal	245	(null)	2649.69	15-APR-26	REF071	ACC01	30-APR-26 11.14.14.861000000 PM
8	319	7	EXT_ONLY	External txn not matched to internal	246	(null)	3732.46	16-APR-26	REF072	ACC01	30-APR-26 11.14.14.861000000 PM
9	320	7	EXT_ONLY	External txn not matched to internal	247	(null)	2807.85	04-APR-26	REF073	ACC02	30-APR-26 11.14.14.861000000 PM
10	321	7	EXT_ONLY	External txn not matched to internal	249	(null)	3673.46	01-APR-26	REF075	ACC03	30-APR-26 11.14.14.861000000 PM
11	322	7	EXT_ONLY	External txn not matched to internal	250	(null)	4389.66	25-APR-26	REF076	ACC05	30-APR-26 11.14.14.861000000 PM
12	323	7	EXT_ONLY	External txn not matched to internal	251	(null)	407.02	22-APR-26	REF077	ACC05	30-APR-26 11.14.14.861000000 PM
13	324	7	EXT_ONLY	External txn not matched to internal	252	(null)	4150.72	19-APR-26	REF078	ACC03	30-APR-26 11.14.14.861000000 PM
14	325	7	EXT_ONLY	External txn not matched to internal	253	(null)	4197.51	20-APR-26	REF079	ACC01	30-APR-26 11.14.14.861000000 PM
15	326	7	EXT_ONLY	External txn not matched to internal	256	(null)	386.98	05-APR-26	REF082	ACC04	30-APR-26 11.14.14.861000000 PM
16	327	7	EXT_ONLY	External txn not matched to internal	257	(null)	3641.89	03-APR-26	REF083	ACC04	30-APR-26 11.14.14.861000000 PM
17	328	7	EXT_ONLY	External txn not matched to internal	258	(null)	3191.24	14-APR-26	REF084	ACC03	30-APR-26 11.14.14.861000000 PM
18	329	7	EXT_ONLY	External txn not matched to internal	259	(null)	2236.17	21-APR-26	REF085	ACC04	30-APR-26 11.14.14.861000000 PM
19	330	7	EXT_ONLY	External txn not matched to internal	175	(null)	2511.52	13-APR-26	REF001	ACC05	30-APR-26 11.14.14.861000000 PM
20	331	7	EXT_ONLY	External txn not matched to internal	176	(null)	3830.51	11-APR-26	REF002	ACC05	30-APR-26 11.14.14.861000000 PM



```
SELECT * FROM recon_metrics WHERE run_id = 7;
```

All Rows Fetched: 1 in 0.68 seconds

	RUN_ID	EXT_TOTAL_COUNT	EXT_TOTAL_AMOUNT	INT_TOTAL_COUNT	INT_TOTAL_AMOUNT	MATCHED_COUNT	MATCHED_AMOUNT	BREAK_COUNT	BREAK_AMOUNT	CREATED_AT
1	7	85	238067.54	4	8498.45	4	8498.45	81	229569.09	30-APR-26 11.14.14.881000000 PM

1. Introduction

1.1 Background

Organizations receive daily transaction statements from multiple external parties (banks, aggregators, brokers). Internal systems also record transactions; however, due to timing, reference differences, missing entries, or duplicates, the two datasets may not align perfectly.

1.2 Problem Statement

Manual matching between external files and internal database records:

- Consumes significant analyst time
- Causes inconsistent results across users
- Increases risk of overlooking high value breaks
- Makes audits harder due to weak traceability

1.3 Project Goal

Build an automated, auditable reconciliation system that:

- Loads daily external Excel files
- Matches external and internal records using database logic
- Produces discrepancies for review
- Generates performance reports automatically

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2. Objectives

2.1 Functional Objectives

- Automated ingestion of external Excel transaction files
- Canonical storage of external transactions in Oracle tables
- Rule-based reconciliation via PL/SQL stored procedures
- Exception generation for unmatched and discrepant records
- Automated Excel reporting with multiple tabs and alerts

2.2 Non-Functional Objectives

- Auditability: every run linked to file name and checksum
- Repeatability: re-run capability without manual cleanup
- Configurability: internal transaction source (table/view) configurable
- Performance: bulk load + set-based SQL joins

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3. System Architecture

3.1 High-Level Flow

1. External Excel file placed in data/input/
2. Python reads Excel, cleans and normalizes fields
3. Python creates a reconciliation run entry in Oracle (recon_run)
4. Python loads external rows into canonical staging (txn_canonical_ext)
5. Python calls PL/SQL procedure recon_pkg.run_reconciliation(run_id, internal_source)
6. PL/SQL writes:
    - Matches (recon_match)
    - Breaks (recon_break)
    - Metrics (recon_metrics)
7. Python extracts results and generates an Excel report in data/output/

3.2 Technology Roles

- **Python:** bridge layer (Excel → DB; DB → Excel reporting)
- **SQL:** data storage, joins, extraction datasets for reports

- **PL/SQL:** reconciliation business logic + audit output generation
- **Excel:** final stakeholder deliverable (dashboard + alert tabs)

3.3 Architecture Diagram (Mermaid)

flowchart LR

```
B -->|bulk insert| C [(Oracle: txn_canonical_ext)]
B -->|insert run log| R [(Oracle: recon_run)]
C --> P [PL/SQL recon_pkg.run_reconciliation]
A -->[External Excel\nDaily File] |Pandas read/clean| B[Python Ingestion]
P --> M [(Oracle: recon_match)]
P --> K [(Oracle: recon_break)]
P --> T [(Oracle: recon_metrics)]
M --> X [Python Report Writer]
K --> X
T --> X
X --> E [Excel Output\nDashboard + Alerts]
```

---

4. Data Model (Database Design)

Core tables:

- recon_run: run metadata (run_date, source_system, file checksum, status)
- txn_canonical_ext: external transactions for the run in canonical format
- recon_rule: rule catalog (extensible)
- recon_match: matched pairs (external ext_txn_id ↔ internal int_txn_id)
- recon_break: discrepancies (e.g., EXT_ONLY, INT_ONLY)
- recon_metrics: run-level KPIs used by the dashboard

How to read the model (in one minute):

- recon_run is the “parent”: every run produces one set of results.
- External input rows live in txn_canonical_ext (many rows per run).
- Internal rows come from a configurable table/view (INTERNAL_TXN_SOURCE) such as internal_txn_sample for demo.
- Matches link one external row to one internal row (recon_match).
- Breaks are exceptions: external-only or internal-only (stored in recon_break).
- Metrics are the dashboard totals for that run (recon_metrics).

Design principles:

- Every result row is tied to a run_id for traceability.
- External rows are hashed per run using row_hash to reduce duplicate ingestion.

- Metrics are stored for fast reporting and trend analysis.

4.1 ER Diagram (Oracle Screenshot)

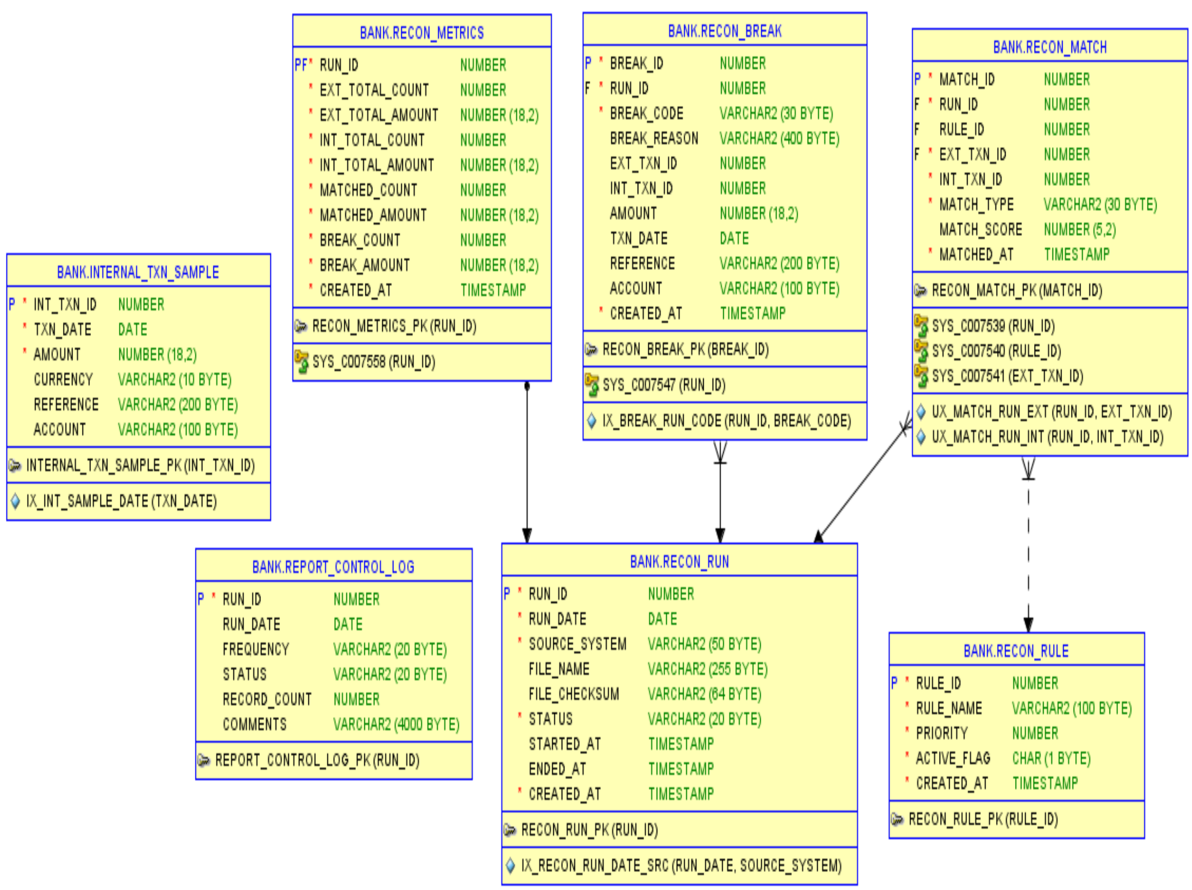
- This is the actual ER diagram generated from the Oracle schema, showing how the tables connect.

4.2 ER Diagram (Mermaid)

RECON_RUN ||--o{ TXN_CANONICAL_EXT : "loads external rows"
RECON_RUN ||--o{ RECON_MATCH : "produces matches"
RECON_RUN ||--o{ RECON_BREAK : "produces breaks"
RECON_RUN ||--|| RECON_METRICS : "summarizes totals"

RECON_RULE ||--o{ RECON_MATCH : "rule used"
TXN_CANONICAL_EXT ||--o{ RECON_MATCH : "matched external row"

INTERNAL_TXN_SOURCE ||--o{ RECON_MATCH : "matched internal row"
INTERNAL_TXN_SOURCE ||--o{ RECON_BREAK : "internal-only break (optional)"
TXN_CANONICAL_EXT ||--o{ RECON_BREAK : "external-only break (optional)"



5. Reconciliation Logic (PL/SQL)

The reconciliation engine is implemented as:

Procedure:

recon_pkg.run_reconciliation(p_run_id, p_internal_source)

5.1 Internal Source

The internal transaction table/view is configurable via environment variable:

- **INTERNAL_TXN_SOURCE=INTERNAL_TXN_SAMPLE** (demo)
- Or a real view/table like
  INTERNAL_TXN_VW
- Required internal columns (current baseline):
  int_txn_id, txn_date, amount, reference, account

5.2 Implemented Rule (Baseline)

Exact 4-key match:

- reference (normalized via TRIM/UPPER)
- amount (exact)
- txn_date (exact)
- account (normalized via TRIM/UPPER)

The procedure enforces 1-to-1 matching so one external row cannot match multiple internal rows and vice versa.

5.3 Break Categories

Generated into recon_break:

- EXT_ONLY: external row not matched to any internal row
- INT_ONLY: internal row not matched to any external row

5.4 Metrics

Stored in recon_metrics:

- External totals (count, sum)
- Internal totals (count, sum)
- Matched totals (count, sum)
- Break totals (count, sum)
- Match rate

6. Reporting Layer (Excel Output)

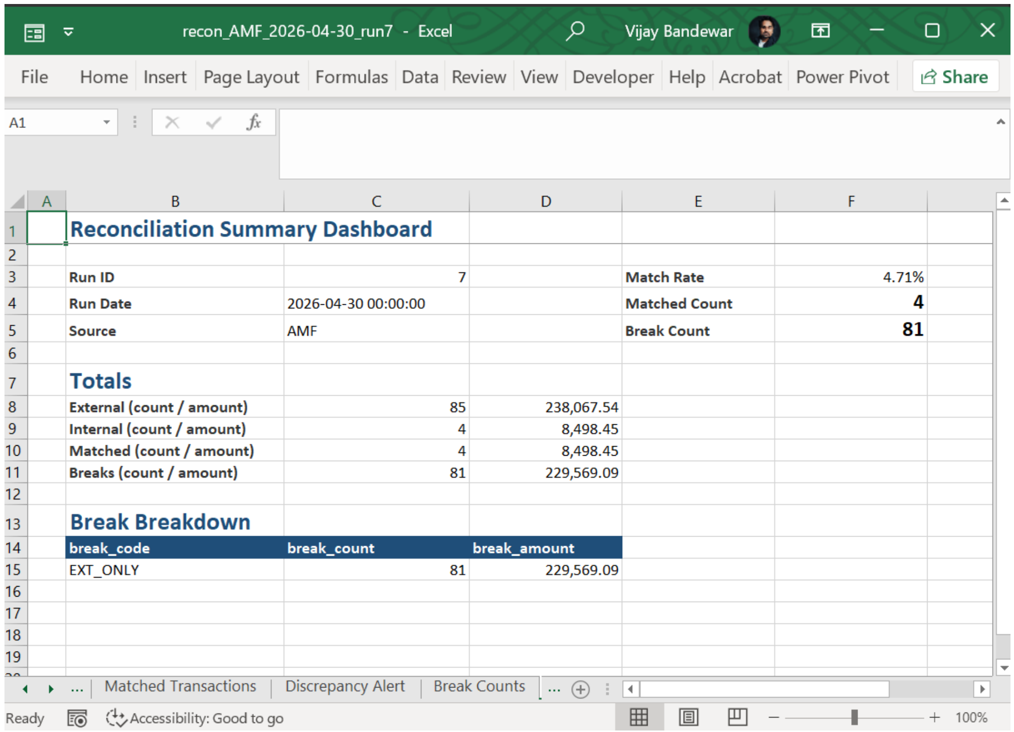
Python generates a multi tab Excel report per run:

Example file:

recon_AMF_2026-04-29_run4.xlsx

6.1 Workbook Tabs

- **Summary Dashboard:** KPIs + totals + break breakdown



Reconciliation Summary Dashboard				
Run ID	7		**Match Rate**	4.71%
Run Date	2026-04-30 00:00:00		**Matched Count**	**4**
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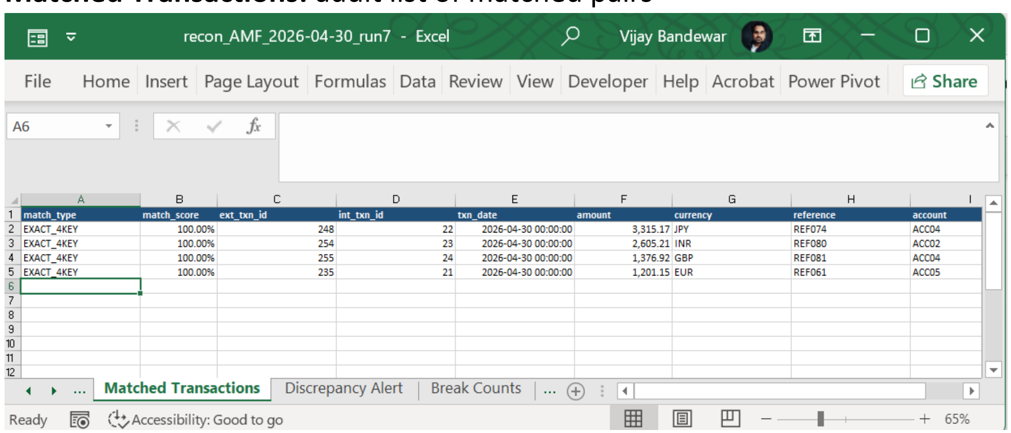
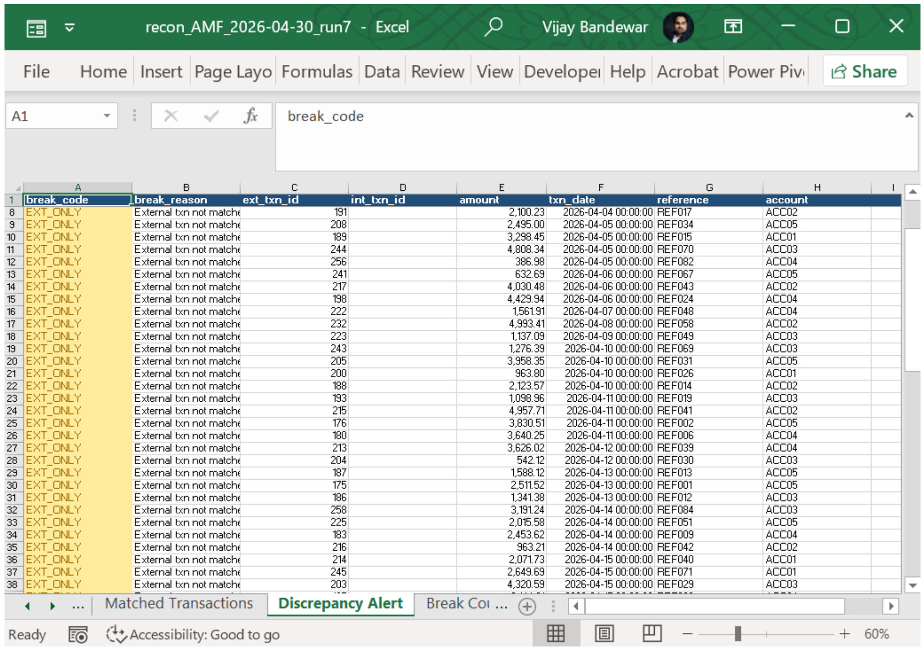
Totals				
External (count / amount)	85	238,067.54		
Internal (count / amount)	4	8,498.45		
Matched (count / amount)	4	8,498.45		
Breaks (count / amount)	81	229,569.09		
Break Breakdown				
break_code	**break_count**	**break_amount**		
EXT_ONLY	81	229,569.09		

- **Matched Transactions:** audit list of matched pairs

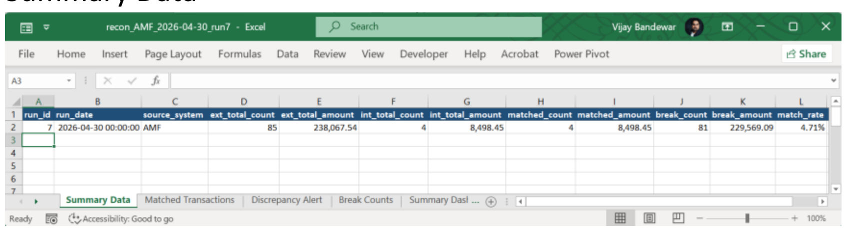


match_type	match_score	ext_txn_id	int_txn_id	txn_date	amount	currency	reference	account
EXACT_4KEY	100.00%	248	22	2026-04-30 00:00:00	3,315.17	JPY	REF074	ACC04
EXACT_4KEY	100.00%	254	23	2026-04-30 00:00:00	2,605.21	INR	REF080	ACC02
EXACT_4KEY	100.00%	255	24	2026-04-30 00:00:00	1,376.92	GBP	REF081	ACC04
EXACT_4KEY	100.00%	235	21	2026-04-30 00:00:00	1,201.15	EUR	REF061	ACC05

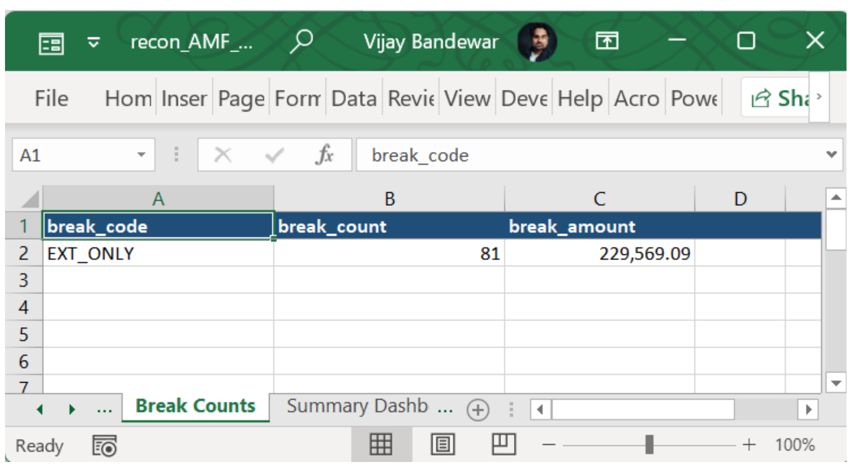
- **Discrepancy Alert:** break list (exceptions) for investigation



- **Supporting tabs:**
    - Summary Data



    - Break Counts



6.2 Conditional Formatting

The system highlights high value discrepancies in the Discrepancy Alert sheet:

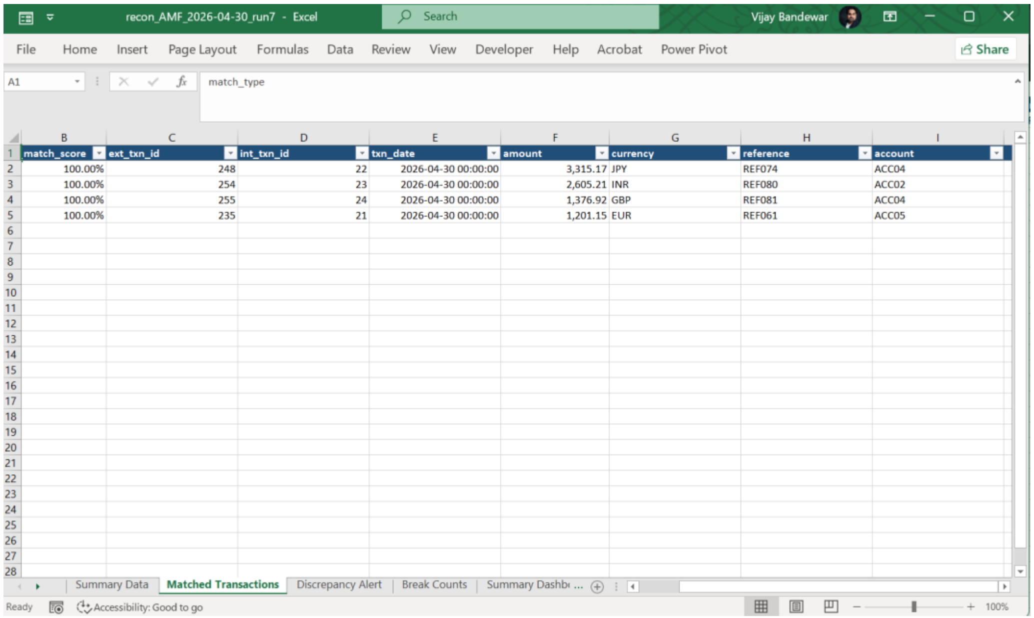
- Highlights when (|amount| \ge) **DISCREPANCY_HIGHLIGHT_AMOUNT**

This allows stakeholders to prioritize the largest breaks first.

6.3 Sample KPI Graphs (What to Include)

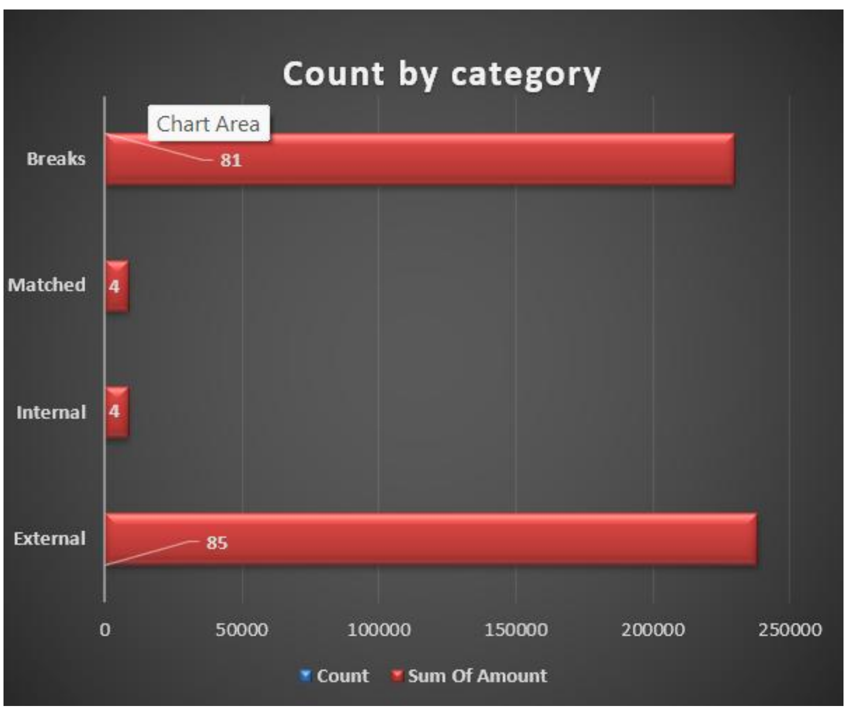
Excel (or screenshots) such as:

- **Match Rate %** by day (line chart)

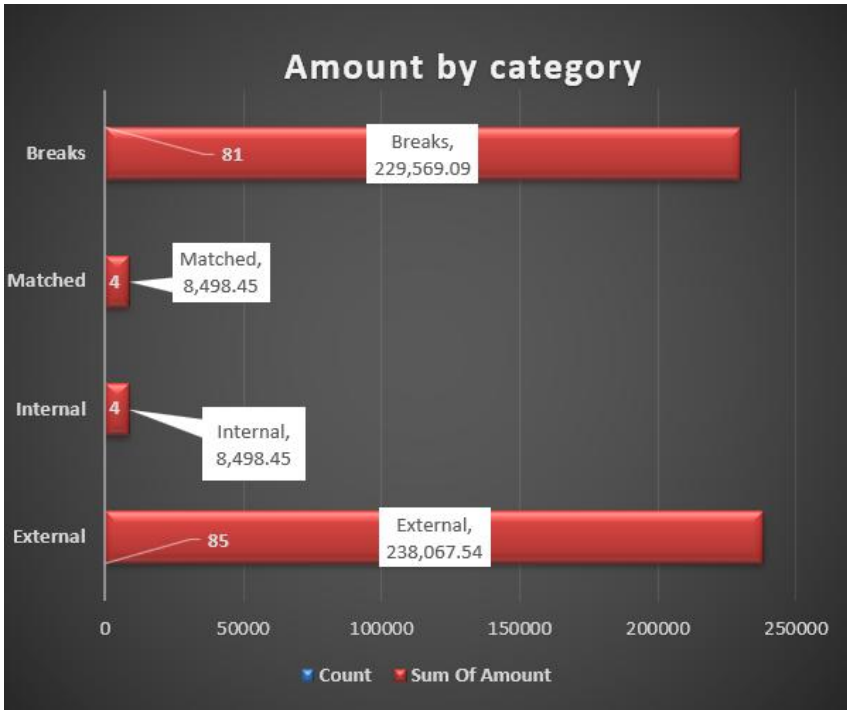


match_score	ext_txn_id	int_txn_id	txn_date	amount	currency	reference	account
100.00%	248	22	2026-04-30 00:00:00	3,315.17	JPY	REF074	ACC04
100.00%	254	23	2026-04-30 00:00:00	2,605.21	INR	REF080	ACC02
100.00%	255	24	2026-04-30 00:00:00	1,376.92	GBP	REF081	ACC04
100.00%	235	21	2026-04-30 00:00:00	1,201.15	EUR	REF061	ACC05

- **Break Count** by category (bar chart: **EXT_ONLY vs INT_ONLY vs others)**



- **Break Amount** by category (stacked bar)



- **Top 10 high-value breaks** (bar chart)

7. Implementation Details (Project Components)

7.1 Python Modules

- **src/excel_ingest.py:**
  reads and normalizes external Excel rows;
  calculates row hashes
- **src/main.py:**
  orchestration
  (create run, load ext rows, call PL/SQL, extract results, write report)
- **src/report_writer.py:**

creates formatted Excel output with multiple tabs and conditional formatting

- **src/db.py:**
  Oracle connection utilities
- **src/config.py:**
  loads. env configuration

7.2 Configuration

In. env (repo root):

- ORACLE_USER, ORACLE_PASSWORD, ORACLE_DSN
- INTERNAL_TXN_SOURCE
- OUTPUT_DIR
- DISCREPANCY_HIGHLIGHT_AMOUNT

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8. Testing / Demonstration

8.1 Test Setup

- Created internal sample table INTERNAL_TXN_SAMPLE
- Inserted sample internal transactions for run date
- Prepared test.xlsx with required columns

8.2 Execution Command

- python -m src.main --source AMF --run-date 2026-04-29 --input "data/input/test.xlsx"

8.3 Expected Outputs

- Oracle populated with recon_run, txn_canonical_ext, recon_match, recon_break, recon_metrics
- Excel report generated in data/output/ with dashboard and alert tabs

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9. Results and Benefits

9.1 Operational Benefits

Reduced manual effort by automating matching and break identification
Consistent outcomes using stored procedure logic
Faster reporting with run metrics and break categories

9.2 Governance / Audit Benefits

File checksum and run logs provide traceability
Repeatable runs (results can be regenerated)

Exceptions are categorized and stored for audit review

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10. Limitations and Future Enhancements

Current baseline uses strict matching. Enhancements to add:
Amount tolerance ((\pm) threshold)
Date window matching ((\pm) N days)
Fuzzy reference matching (normalized text scoring)
One-to-many matching for batch settlements/fees
Automated email distribution and archival
Trend dashboards (Power BI) from recon_metrics and recon_break

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11. Conclusion

The Automated Multi Source Financial Reconciliation System provides an auditable, repeatable, and business-friendly reconciliation workflow. It integrates Python-based ingestion, Oracle-based reconciliation logic, and Excel-based reporting to reduce manual work and improve accuracy in daily reconciliation operations.